The proof of equation (2.5.2) from (2.5.1) is similar to the proof of section 1.4.1 (Ch. 1, p.5).

$$S_T^{[2]} = \alpha S_T + (1-\alpha) S_{T-1}^{[2]} = \alpha S_T + \beta S_{T-1}^{[2]} \qquad (2.5.1)$$

$$[\ \beta = 1-\alpha\ ]$$

$$S_{T-1}^{[2]} = \alpha S_{T-1} + \beta S_{T-2}^{[2]}$$

$$S_{T-2}^{[2]} = \alpha S_{T-2} + \beta S_{T-3}^{[2]}$$

$$\vdots$$

$$S_1^{[2]} = \alpha S_1 + \beta S_0^{[2]}$$

By substitution,

$$S_T^{[2]} = \alpha S_T + \beta\left[\alpha S_{T-1} + \beta S_{T-2}^{[2]}\right]$$

$$= \alpha S_T + \alpha\beta S_{T-1} + \beta^2 S_{T-2}^{[2]}$$

$$= \alpha S_T + \alpha\beta S_{T-1} + \alpha\beta^2 S_{T-2} + \beta^3 S_{T-3}^{[2]}$$

$$\vdots$$

$$S_T^{[2]} = \alpha \sum_{k=0}^{T-1} \beta^k S_{T-k} + \beta^T S_0^{[2]}$$

As $T \to \infty$, then $\lim_{T\to\infty} \beta^T S_0^{[2]} = 0$

Hence,

$$E\left[S_T^{[2]}\right] = E\left[\alpha \sum_{k=0}^{\infty} \beta^k S_{T-k}\right] = \alpha \sum_{k=0}^{\infty} \beta^k E(S_{T-k})$$

$$= \alpha \sum_{k=0}^{\infty} \beta^k \left[b_1 + b_2(T-k)\right] = \alpha \sum_{k=0}^{\infty} \beta^k \left[b_1 + b_2 T - b_2 k\right]$$

$$= \alpha(b_1 + b_2 T) \sum_{k=0}^{\infty} \beta^k - \alpha b_2 \sum_{k=0}^{\infty} k\beta^k$$

$$= \alpha(b_1 + b_2 T) \cdot \frac{1}{1-(1-\alpha)} - \alpha b_2 \cdot \frac{\beta}{\alpha^2}$$

$$= b_1 + b_2 T - \frac{\beta}{\alpha} b_2$$

$$= E(S_T) - \frac{\beta}{\alpha} b_2 \qquad (2.5.2)$$